CARD G · GLAZE RECIPE SEARCH · TEST RESULTS

Glaze Search Evaluation

Every test of the glaze recipe search, stage by stage: the data and labels it learns from, the photo input, the text reader, the trained models, and the final ranking. All model scores come from held-out test recipes that training never saw.

Measured **5 Nov 2026** Smart search model **`78afd7dfbe2c`** Colour-match model **`70cd01c0c673`**

Do data gathering and self-labelling count?

Yes, as their own layer: data and label quality, reported apart from model accuracy.

Several tests score the models against answer keys the project made itself: glaze colours measured from photos, attribute labels written by an AI from the images, and typed sentences generated from recipe facts. A model can only be as right as its answer key. Report how good each key is next to the scores that depend on it, and don't present the self-made labels as ground truth.

Who made each answer key

ANSWER KEY	MADE BY	USED TO SCORE	CHECKED BY A PERSON?
Recipe identity (is this the recipe's own record?)	Glazy database	R@K of both models	reliable by definition
Oxide formula, cone, surface, transparency, atmosphere	Potters on Glazy	P@10, text reader, tag suggestions, MAD@K	no, crowd-entered
Glaze colour (hex + colour word)	Our pipeline: segmentation + ColorThief on the photo	Colour score, P@10, text reader colour, colour-match training text	no
Object cut-outs (masks)	Grounding DINO + SAM 2.1, checked by CLIP	Every photo colour and photo embedding	yes: reviewer notes, before/after test, 2 review rounds

ANSWER KEY	MADE BY	USED TO SCORE	CHECKED BY A PERSON?
Attributes: form, texture, finish, effects, defects	Claude, from the object image only	Photo attribute model (card 1)	no
Typed test sentences with typos	Generated from each recipe's facts	Smart search R@K, P@10, text reader	synthetic, not real users

Evaluation map

One row per stage of the "How it works" diagram. Status: measured partly not measured

STAGE	WHAT IS CHECKED	METRICS	HEADLINE RESULT	STATUS
0	Data: coverage, duplicates, odd records	% filled, duplicate groups, plausibility checks	99.8 % have a photo and a formula; 11.9 % of recipes have a near-copy	measured
0	Self-made colour labels	Consistency check now; label precision needs a human sample	80 % of same-chemistry pairs carry different colour names	partly
A	Photo → object cut-out (pretrained models vs reviewer notes)	Before/after verdicts on held-out photos, IoU with the old mask, review rounds	148 of 168 reviewer-flagged photos improved; 2 review rounds cleaned 1,665 photos	measured
A	Photo attributes (card 1)	Accuracy vs most-common guess, coverage at 98 % precision	Form 83.6 % vs 55.5 %; scored against AI labels	partly
B	Text reader (words → values)	% correct when a value is read	Cone 99.3 %, colour 93.3 %, surface 89.1 %	measured
B	Tag suggestions	Accuracy, macro-F1, most-common baseline	Colour family no better than guessing	measured
C · D	Smart search model (text, photo, formula)	R@1/5/10, median rank, zero-shot vs trained	Typed sentence R@10 74.2 % (untrained 62.0 %)	measured
C · D	Colour-match model (text ↔ chemistry)	R@K, right vs wrong pair cosine, MAD@K	R@10 8.2 % (random 0.8 %)	measured
E	Full ranking (all parts + weights)	P@10 with automatic relevance	73.1 % with text model + read values	measured
F	Results in use	Speed; human relevance (nDCG@10) not yet	Smart search median 23 ms	partly

STAGE	WHAT IS CHECKED	METRICS	HEADLINE RESULT	STATUS
C · D · E	Text + formula given together	P@10 with the recipe's own group left out	–	not measured

At a glance

73.1 %

of the top 10 are relevant for a typed sentence, with text model and read values combined

text model alone 54.4 % · read values alone 55.3 %

74.2 %

of misspelled sentences find their recipe in the top 10 (smart search)

MiniLM alone, untrained 62.0 % · random 0.9 %

8.2 %

of recipe texts find their exact chemistry in the top 10 (colour match)

random 0.8 % · about 10× random

99.3 %

of cones read from a typed sentence are correct

colour 93.3 % · surface 89.1 % · atmosphere 100 %

148 / 168

reviewer-flagged photos were cut out better by the pretrained-model segmentation

fresh normal photos: 57 of 60 same or better

How it was tested

Smart search (text + photo + formula)

- 7,063 recipes, each with a photo, a 26-oxide formula and tags.
- Recipes are grouped so copies stay together: same formula (rounded to 0.1), a near-identical photo, or the same name by the same author. That gives 6,537 groups.
- Groups are split 70 / 15 / 15: 4,945 train, 1,059 validation, 1,059 test.
- Validation picks the best epoch. Test is used only once, at the end.

Colour match (text ↔ chemistry)

- 6,718 recipes. Groups: recipes whose √-fraction chemistry is within cosine distance 0.01 (2,475 groups, largest 47).
- Groups are split 80 / 20: 1,309 test recipes. Training recipes almost identical to a test recipe (cosine > 0.99) are removed: 2,529 dropped, 2,880 left for training.
- Leak check: no test recipe has a training recipe above cosine 0.99 (99.1 % have one above 0.95).
- The notebook's own test code was re-run on the shipped model. The rebuilt split gives exactly the same 1,309 test recipes as the shipped catalog.

Every metric used on this page

R@K	Share of queries whose own recipe is in the top K results. In the smart search a recipe from the same group also counts as found. Higher is better.
Random R@10	What blind guessing would score: 10 ÷ number of recipes searched.
Median rank	The middle position of the right recipe across all queries. 1 = first place. Lower is better.
P@10	Share of the top 10 that are relevant. Relevant means the colour is within ΔE_{00} 20 of the query recipe's colour, with the same surface and an overlapping cone where those are known.
Right vs wrong pair cosine	Average similarity of a text with its own chemistry, against its similarity with every other recipe's chemistry. A bigger gap means better alignment.
MAD@K	Mean absolute difference of the 26 oxide fractions between a query recipe and its top K results. Lower means closer chemistry.
ΔE_{00}	CIEDE2000 colour difference. About 2 is just visible, and above 20 is a clearly different colour.
Accuracy · baseline	Share of correct predictions, next to always guessing the most common answer. A model has to beat that baseline to be useful.

Macro-F1	F1 averaged over classes, so rare classes count as much as common ones.
Coverage at 98 % precision	The attribute model answers only when its confidence passes a cut-off that reached 98 % precision on validation. Coverage is the share it answers; Wilson low is the 95 % lower bound of its precision on test.
Speed	Server time per search from the app's own search log.
Before / after verdict	The old and new cut-out of the same photo side by side, judged improved, same, worse, both bad or unclear.
AUROC	How well a score separates photos with an effect from photos without it. 0.5 is chance, 1.0 is perfect.
Proposed: IoU, nDCG@10, Cohen's κ	Mask overlap with a hand-drawn mask; ranking quality from graded human judgements; agreement between a person and the AI labels. See "Limits to state in the report".

0 Data and labels

0a · Coverage of the 7,063 recipes

FIELD	RECIPES	SHARE	SOURCE
Photo with measured colours	7,052	99.8 %	Glazy photo, our segmentation
Oxide formula usable by the model	7,049	99.8 %	Glazy
Cone	6,383	90.4 %	Glazy
Atmosphere	6,127	86.7 %	Glazy
Written description	4,766	67.5 %	Glazy
Surface	4,008	56.7 %	Glazy
Transparency	3,968	56.2 %	Glazy

Surface and transparency are missing for about 44 % of recipes. Those recipes can't be checked on surface in P@10, and the text reader has nothing to compare with.

0b · Duplicates and odd records

CHECK	RESULT	WHAT IT MEANS
Recipes with a near-copy (same formula, near-identical photo, or same name and author)	841 · 11.9 %	Kept in the same group so a copy can't leak from training into test. 6,537 groups, the largest has 17 recipes.
Formulas with less than 20 % silica	60	Add-on lists rather than full glazes (e.g. "TinTan": 5 titanium + 1 tin). Their chemistry can't explain the photo.
Recipes using a commercial stain (colour-match catalog)	316 · 4.7 %	The stain is not in the 26 oxide columns, so the colour comes from something the model can't see.

0c · Self-made colour labels

Each recipe's colour is measured by ColorThief on its segmented photo. Nobody has checked these labels by hand yet. A consistency check shows how noisy they are: recipes with near-identical chemistry should mostly look alike.

RECIPE PAIRS WITH NEAR-IDENTICAL CHEMISTRY (COSINE > 0.999)	5,486 PAIRS
Different colour name	80.3 %
Different colour family (blue, green …)	58.2 %
Colour differs by $\Delta E_{00} > 20$	30.2 %
Median colour difference (random pairs: 25.9)	ΔE 12.5

Part of this is real: the same glaze looks different on another clay, at another thickness or firing. Part is label error. Spot checks found labels taken from the background (a green glaze labelled "light gray") and recipes whose colour comes from a missing stain.

A Photo input: object cut-outs and attributes

A1 · Segmentation upgrade (Grounding DINO + SAM 2.1 + CLIP) judged against the human review

Reviewers first marked the old pipeline's errors on 343 photos. 235 got an error label: whole frame boxed 139, inner part missed 52, outer part missed 13, objects merged 13, OK 14, other 4. The photos were split before development: 175 to design and tune, 168 held out. Two samples of normal photos checked that nothing else broke; the second one (60 photos) was never used for any decision.

PHOTO SET	N	IMPROVED	SAME	WORSE	BOTH BAD	UNCLEAR
Reviewer-flagged photos, held-out half	168	148	9	2	7	2
Normal photos, hold-out sample	48	10	38	0	0	0

PHOTO SET	N	IMPROVED	SAME	WORSE	BOTH BAD	UNCLEAR
Normal photos, fresh sample (never used for any decision)	60	14	43	3	0	0

Automatic checks on the same photos: in 96 of 107 labelled error photos the new mask changed the way the reviewer's label calls for. On normal photos, 42 of 43 (hold-out) and 47 of 60 (fresh) keep IoU ≥ 0.85 with the old mask, and most of the rest are improvements. Verdicts were made by eye on before/after sheets; no pixel ground truth exists. 8 hold-out photos were looked at again after the first pass, so the fresh sample is the cleanest test. Its 3 regressions are known and not fixed.

A2 · Review rounds after the full run (review app)

STEP	RESULT
Cut-outs made from 7,052 photos	14,940
Accepted automatically · flagged "review required" by the pipeline	12,194 · 2,746 (18.4 %)
Photos checked by reviewers (two rounds, 15–16 Oct 2026)	1,665 (23.6 %)
Cut-outs in those photos	6,772
Removed by reviewers as not needed	3,010
Corrected by hand (merged, split, re-cut, missing objects added)	695 in 308 photos
Cut-outs used by the search after review	11,930

These rounds cleaned the data the search uses. The pipeline's own "review required" flag did not predict which cut-outs reviewers removed (40.4 % of flagged vs 46.1 % of auto-accepted, in the reviewed photos), so it is a weak guide for choosing what to review. "Not needed" also covers duplicate or unimportant pieces, not only bad masks.

A3 · Photo attribute model (card 1), scored against AI labels

Labels written by Claude from each object image; not checked by a person. These numbers show agreement with those labels, not with the truth. Test set: 240 objects (train 1,504, validation 240).

ATTRIBUTE	CLASSES	TEST ACCURACY	MOST-COMMON GUESS	ANSWERED AT THE 98 % CUT-OFF
Form	11	83.6 %	55.5 %	34.0 % answered, precision 100 % (Wilson low 95.5 %)
Texture	4	85.0 %	85.0 %	48.8 % answered, precision 96.6 % (low 91.5 %)
Transparency	4	74.6 %	73.3 %	no cut-off reached 98 %, so always "uncertain"
Transparency group	2	81.3 %	80.0 %	always "uncertain"

ATTRIBUTE	CLASSES	TEST ACCURACY	MOST-COMMON GUESS	ANSWERED AT THE 98 % CUT-OFF
Surface group	3	63.8 %	53.8 %	always "uncertain"
Surface finish	9	45.4 %	42.9 %	always "uncertain"
Glaze effect (several per object)	8	R@1 49.4 %	38.5 %	157 objects have an effect
Defect (several per object)	5	R@1 50.0 %	27.3 %	only 11 labelled objects: too few to judge

A4 · Glaze-effect detectors (CLIP prompts), scored against the recipe subtype

Weak labels: the recipe's Glazy subtype names an intended effect. 306 positive and 300 random photos, split by recipe; thresholds chosen on the development half only. AUROC on the test half, old single prompt vs new prompt ensemble:

EFFECT	OLD	NEW
Pitted (crater)	0.60	0.84
Crawling	0.58	0.77
Oil spot	0.73	0.86
Crystalline	0.68	0.76
Metallic	0.78	0.83
Crackle	0.71	0.67

Effects are rare (0.45–1.7 % of photos), so the expected precision at real prevalence is low (about 2–25 %). Every predicted effect or defect is flagged for review.

B Text reader and tag suggestions

B1 · Values read from a typed sentence

VALUE	SENTENCES WHERE IT WAS READ	CORRECT	CORRECT MEANS
Colour	1,023	93.3 %	within ΔE_{00} 20 of the recipe's measured colour
Cone	667	99.3 %	overlaps the recipe's Orton cone range
Surface	503	89.1 %	same surface word
Atmosphere	611	100 %	one of the recipe's atmospheres

1,059 generated test sentences with a 15 % typo rate, e.g. `pls medium graay specialty crackkle gas reduction`. The reader is rules plus spelling correction;

no training is involved.

B2 · Tag suggestions from free text (TF-IDF + logistic regression)

TAG	N	ACCURACY	MOST-COMMON GUESS	MACRO-F1	VERDICT
Surface (9 classes)	453	63.4 %	52.8 % (glossy)	0.41	better than guessing
Transparency (4)	448	67.9 %	47.8 % (opaque)	0.64	better than guessing
Atmosphere (6)	668	92.7 %	87.9 % (oxidation)	0.61	small gain
Colour family (12)	712	36.7 %	37.4 % (gray)	0.27	no better than guessing

Colour family is the one tag whose answer key is self-made (from the photo colour), and it is the one that fails. The app shows suggestions only as hints; the search uses only the values from B1.

C · D Frozen encoders and trained models

D1 · Smart search: finding the right recipe from one kind of input

Frozen MiniLM (text), DINOv2 (photo) and √ oxide fractions (formula), each with a small trained head into one 256-number space. Test: 1,059 recipes searched, random R@10 = 0.9 %.

R@10 on test random (0.9 %)

QUERY → WHAT IT IS MATCHED AGAINST	N	R@1	R@5	R@10	MEDIAN RANK	VAL R@10
Tag text → recipe's photo + formula "color: moss green, Cone 6, Glossy …"	1,059	9.9 %	28.6 %	40.2 %	16	39.8 %
Noisy text → recipe's photo + formula typed sentence with typos	1,059	7.2 %	18.1 %	28.2 %	33	27.9 %
Recipe description → recipe's photo + formula only recipes that have a written description	713	7.7 %	23.8 %	33.4 %	31	31.3 %
Photo → recipe's text + formula	1,058	14.3 %	34.9 %	45.3 %	14	45.8 %
Formula → recipe's text + photo	1,058	6.6 %	23.7 %	33.1 %	26	34.5 %

Each query is matched only against the recipe's *other* parts, so these numbers measure text ↔ formula ↔ photo across kinds. Validation and test agree within 1.5 points, so the model was not tuned to the test set.

D2 · Untrained (zero-shot) vs trained, for typed sentences

METHOD (NOISY TEST SENTENCE)	R@1	R@5	R@10	MEDIAN RANK
Pretrained MiniLM only, no training: sentence vs each recipe's tag text	24.6 %	50.1 %	62.0 %	5
Trained heads, all parts: sentence vs recipe's text + photo + formula	31.7 %	61.9 %	74.2 %	3

Same 1,059 test sentences. Training adds 12.2 points of R@10 over the frozen text model alone.

D3 · Smart search training

- Symmetric InfoNCE over text–photo, text–formula and photo–formula pairs; recipes of the same group are not used as negatives. Hidden 512, output 256, dropout 0.1, Adam lr 0.001, weight decay 0.0001, batch 256, temperature 0.07, seed 1.
- Early stopping (patience 12): ran 70 epochs, best at **epoch 58** (validation mean R@10 = 36.1 %).
- Trained weights: text head 328,448 · photo head 656,128 · formula head 145,152. All pretrained encoders stay frozen.
- "Explore formulations" families: K-Means, k = 10 chosen by silhouette (0.102, a weak cluster structure). HDBSCAN found only 2 clusters with 35 % noise, so it was not used.

D4 · Colour match: the notebook's own test (1,309 held-out recipes, same-group recipes excluded)

Frozen MiniLM text (384 numbers) → trained linear layer to 256. Chemistry: 26 oxides → fractions → √ → PCA-16 (93.5 % of variance) → trained linear layer to 256.

R@10 random (0.8 %)

MODEL	RIGHT PAIR COS	WRONG PAIRS COS	R@1	R@5	R@10
Trained (shipped model)	0.238	0.076	2.1 %	5.4 %	8.2 %
Same two layers, random weights	0.036	0.037	0.0 %	0.3 %	0.6 %
Random guessing	–	–	0.08 %	0.38 %	0.77 %

D5 · Colour match: how close the retrieved chemistry and colour are

CHECK (1,309 TEST RECIPES)	TRAINED	RANDOM
MAD@1 · MAD@5 · MAD@10, text → chemistry (lower is better)	0.0150 · 0.0152 · 0.0153	0.0185

CHECK (1,309 TEST RECIPES)	TRAINED	RANDOM
MAD@1 · MAD@5 · MAD@10, chemistry → text	0.0165 · 0.0166 · 0.0165	0.0185
Median colour distance of the top 10 to the query recipe	ΔE 22.6	ΔE 27.7
Top 10 in the same colour family (blue, green …)	29.1 %	21.0 %
Top 10 with the same exact colour name	8.6 %	3.7 %

Recipes found from a text have about 17 % closer chemistry than random. The model learns what kind of glaze a text describes better than the exact recipe; the app's 40 % colour-distance part supplies the precise colour.

D6 · Colour match: the app runs the model unchanged, and its limits

- Weights, recipe vectors and catalog in the app are byte-identical (SHA-256) to the delivered model. Parity tests on the notebook's 5 example searches give the same top 10 and "explore" lists, with scores within 0.00001; all 3 pass.
- One deliberate change: the app reads Orton cones (cone 04 ≠ cone 4). This affects only the 10 % cone part of the score.
- **Cobalt and chrome are nearly invisible to it.** PCA-16 keeps 19 % of the cobalt (CoO) variation and 5 % of the chrome (Cr_2O_3); iron (99.7 %) and copper (98 %) are kept.

  Removing all cobalt from a recipe barely moves its vector (median cosine 0.984 over 984 cobalt recipes).

E · F Final ranking and results in use

E1 · Full smart search (the app's ranking) on typed sentences

SCORING USED	QUERIES	P@10
Text model only	300	54.4 %
Read values only (colour, cone, surface … from the text reader)	300	55.3 %
Both (what the app uses)	300	73.1 %

300 generated sentences from random test recipes. Weights (renormalised over the parts a query has): text 0.35, photo 0.35, formula 0.25, colour ΔE 0.30, cone 0.10, surface 0.10, transparency 0.05, atmosphere 0.05. The colour-match score is 0.40 learned match + 0.40 colour ΔE + 0.10 surface + 0.10 cone. Relevance here uses the self-made photo colour (see stage 0).

F1 · Results in use

CHECK	RESULT
Searches logged in the app (3–5 Nov 2026)	66
Smart search speed: median · 90th percentile (52 searches)	23 ms · 65 ms
Colour match speed: median (5 searches)	126 ms
Logged searches with a human relevance judgement	0

Speed is server time per search and excludes cutting objects out of an uploaded photo. The object search (cards 1 and 1b) already shows Hit@K, Precision@K and nDCG for every search and keeps a session evaluation. Card G logs its searches but does not evaluate them yet.

Limits to state in the report, and optional extra checks

None of these blocks the report. They limit what the numbers can claim: state them as limits, and add a check only if there is time.

GAP	WHY IT MATTERS	HOW TO MEASURE	METRIC
0 Colour labels	They are the answer key for the colour score, P@10 and the colour-match text	A person compares the label chip with the glaze in 100 random photos	Label precision; ΔE_{00} to a hand-picked colour
A Pixel-level mask accuracy	The before/after test is judged by eye; there are no hand-drawn masks	Hand-draw masks for about 50 random photos	IoU, boundary F-score
A AI attribute labels	The attribute model is scored only against these labels	A person checks 100 of Claude's labels	Agreement %, Cohen's κ
F Human relevance	Automatic P@10 is not a potter's opinion	30 real text queries; grade each top-10 result 0 / 1 / 2	nDCG@10, human P@10
C · D · E Text + formula together	The app accepts both, but no test gives both	Query = a test recipe's text + formula, with that recipe's group left out of the results	P@10 (same relevance rule)
C · D One comparison of both models	Different splits (1,059 vs 1,309 test recipes) and grouping rules	Score both models on the colour-match 1,309 test recipes	R@10, P@10

- **Keep two counting rules apart in the report.** Smart search R@K counts a same-group recipe as found; the colour-match test removes same-group recipes from the list.

- **Test queries are generated, not typed by users.** Real queries may be shorter, vaguer or in other words.

Related pages: Glaze Search Mind Map (each step and its pretrained model) · Glaze Search Manual (how to use it) · the app online.

Sources: the smart search model's saved test results, the colour-match notebook's own report functions re-run read-only on the shipped model, the attribute model's report, the segmentation upgrade's design and evaluation notes, the segmentation review file, the recipe bank, and the app's search log (read only). Stage 0c and D5–D6 are extra checks on the same test recipes.